# Efficient sequential learning and counterfactual policy learning

Houssam Zenati

Inria Saclay (Mind)

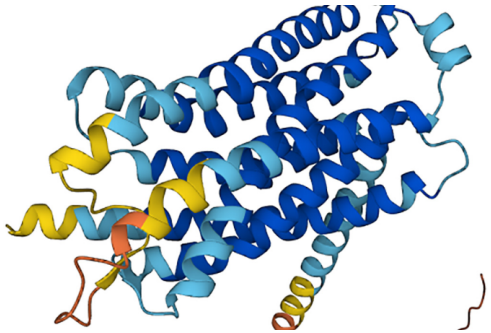
November 13, 2023

*Gatsby reading group, University College London*

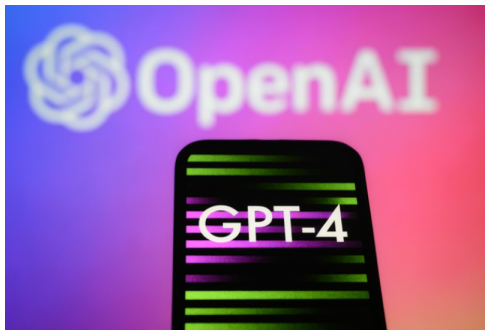
# Successes of artificial intelligence




AlphaGo (2016)



AlphaFold (2018)




ChatGPT (2022)

We often use machine learning/ artificial intelligence to make **decisions**, not only **predictions**.

News feed

Online Retailing

Entertainment

Autonomous driving

# Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



**Context** $x$ (user info) **Policy** $\pi$ selects **action** $a$ **Feedback** $y$

This setting is ubiquitous in many applications.

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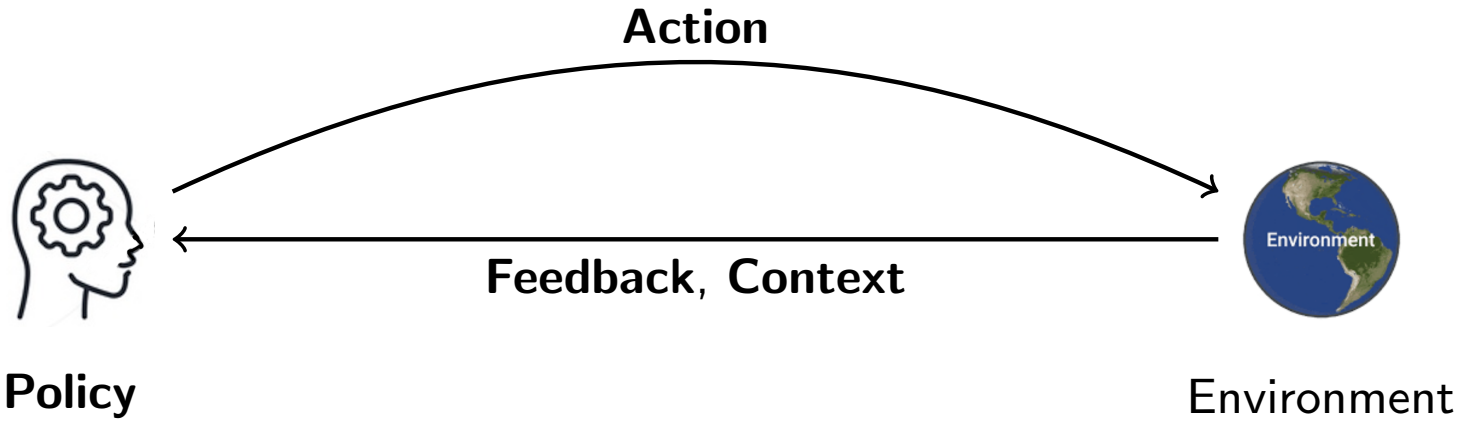
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- precision medicine (Pharma Techs)

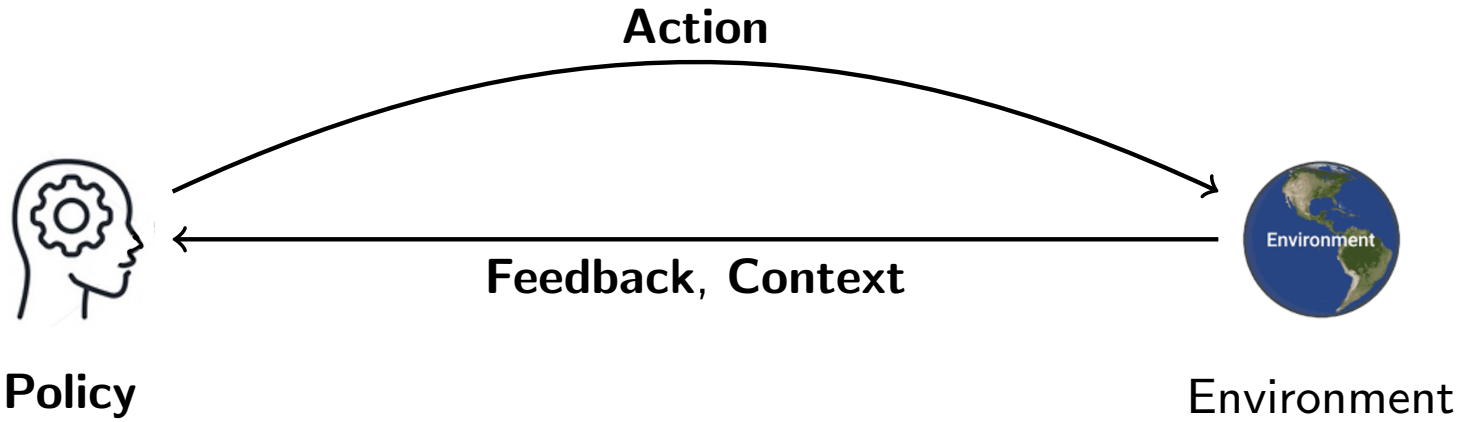
# Offline versus online Learning

Online agents learn by **interacting directly** with the environment.

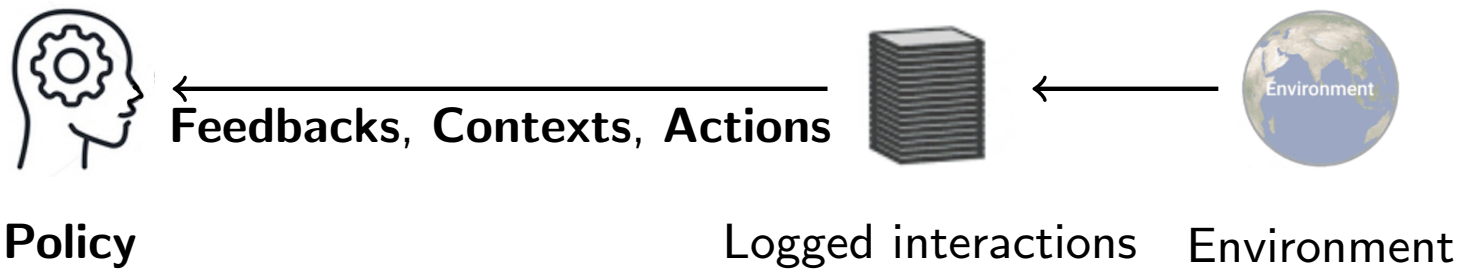


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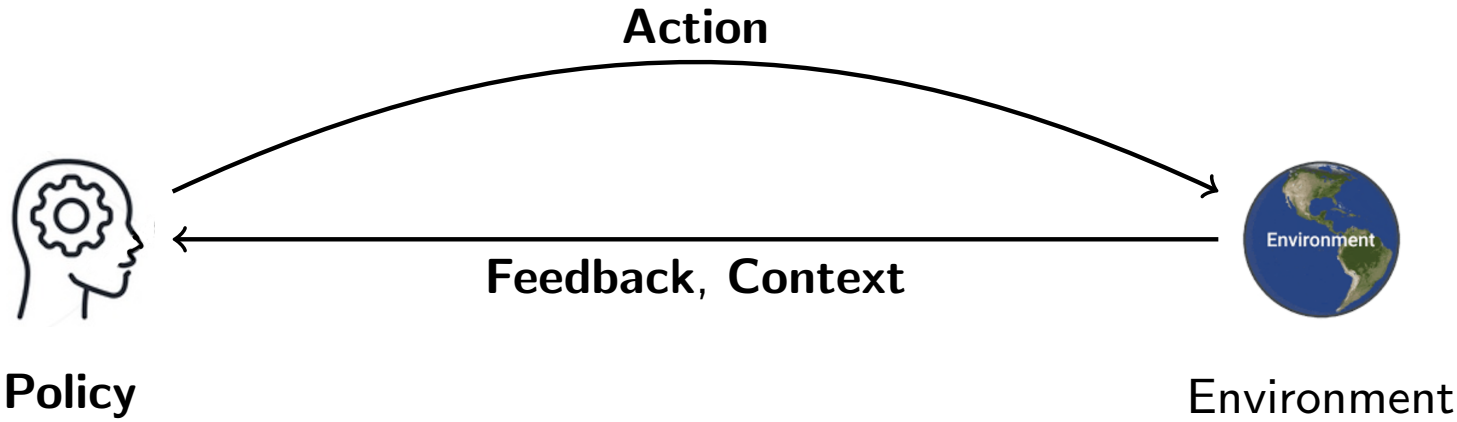



Offline learning methods **leverage past** "logged" **interactions**.

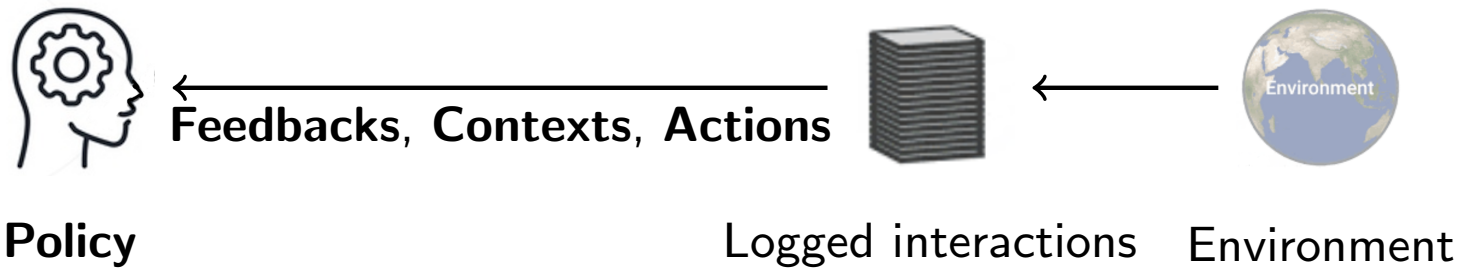


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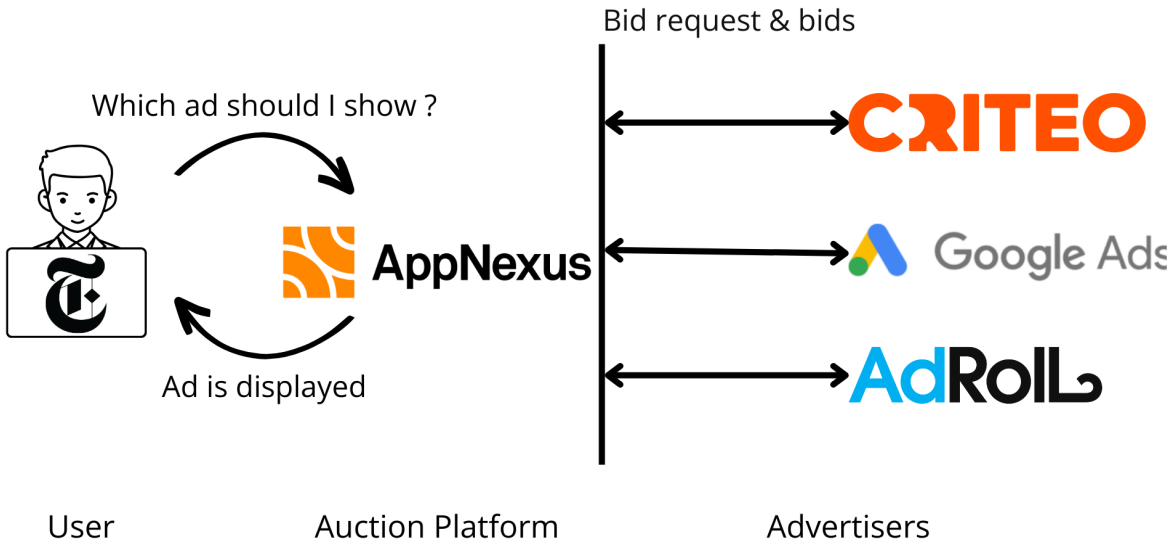

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In real-life applications such as advertising, personalized medicine, offline learning may be preferred because of **exploration** or **safety** constraints.

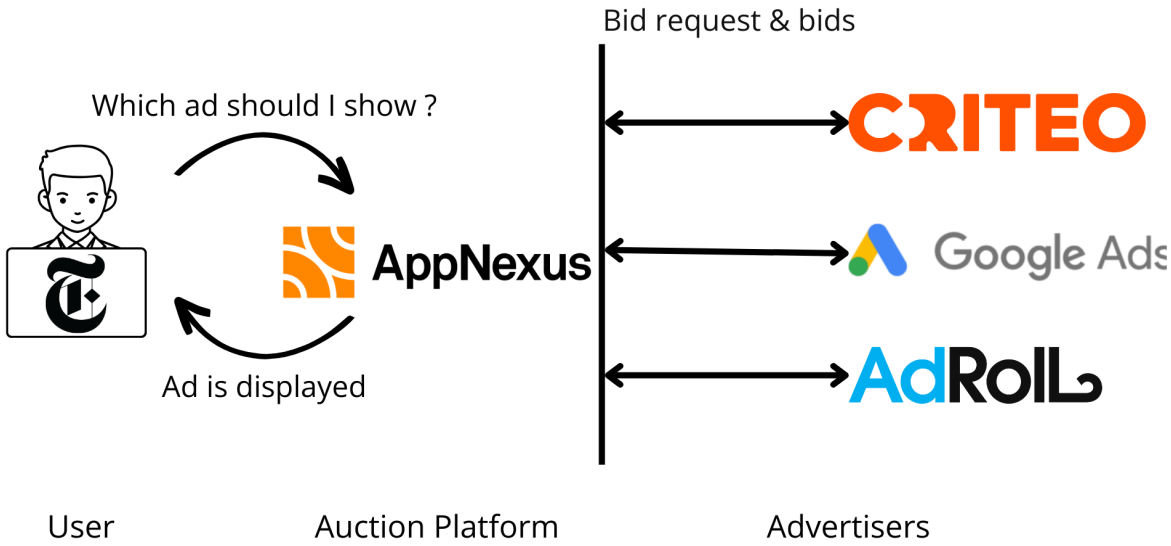
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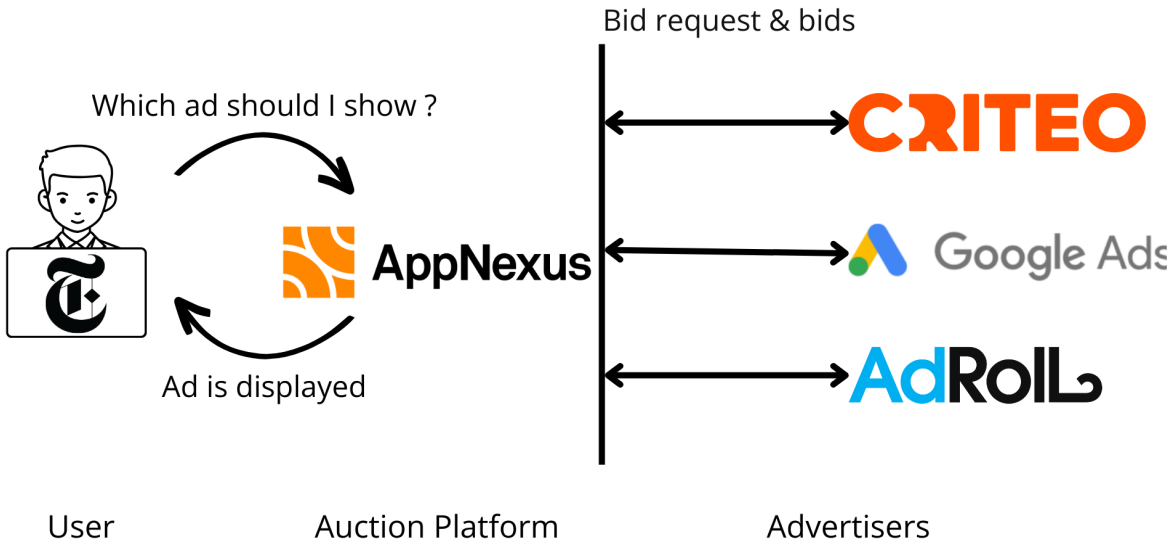
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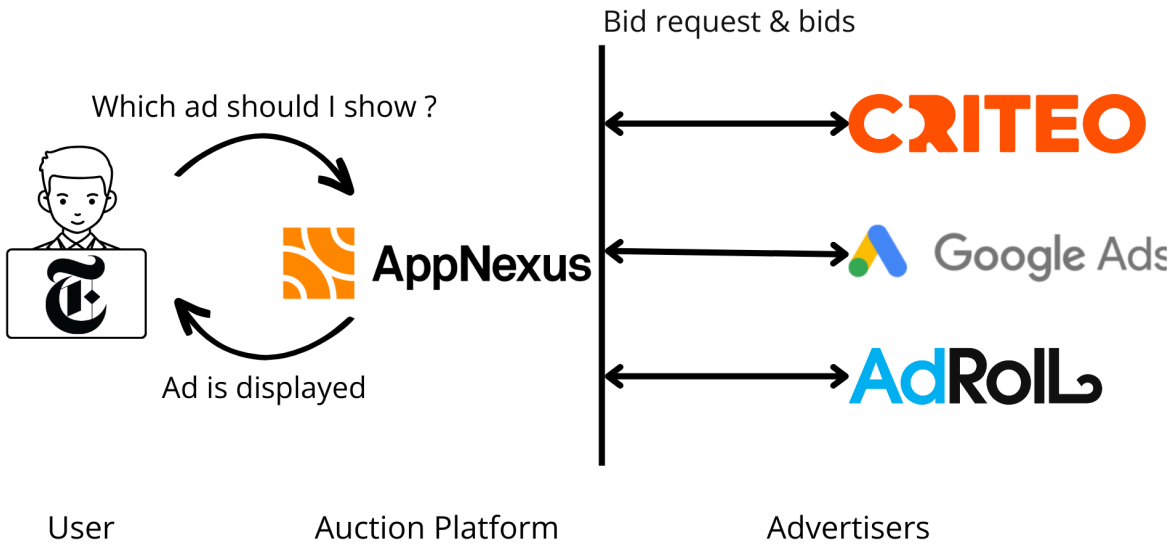
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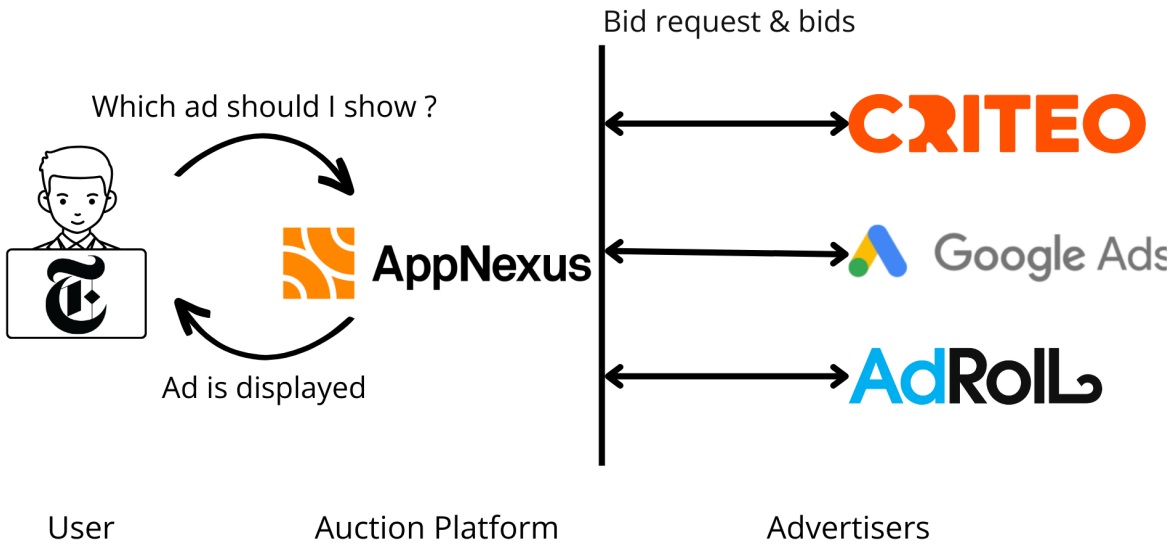
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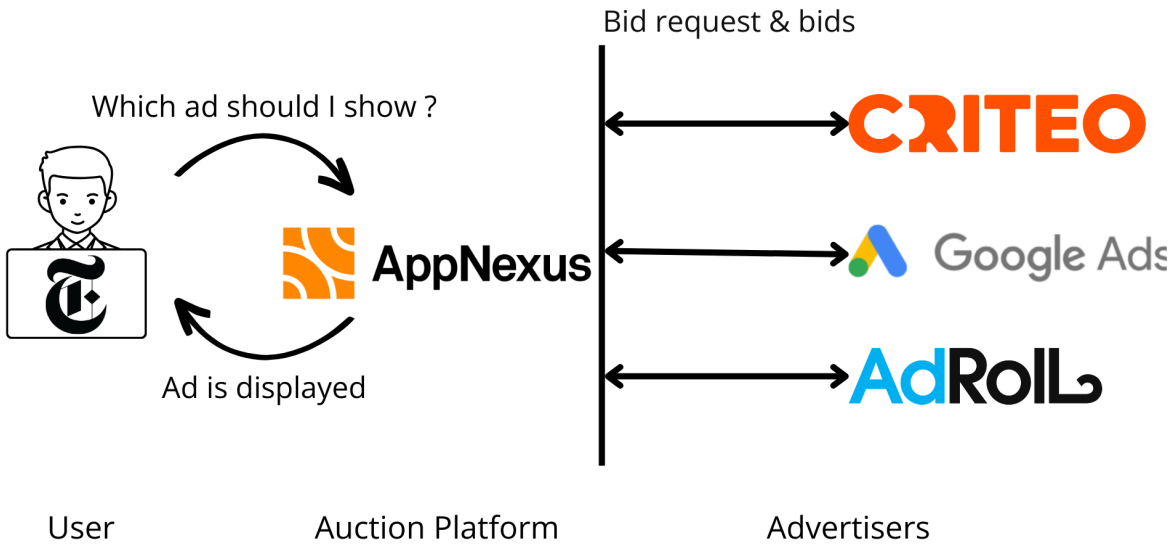

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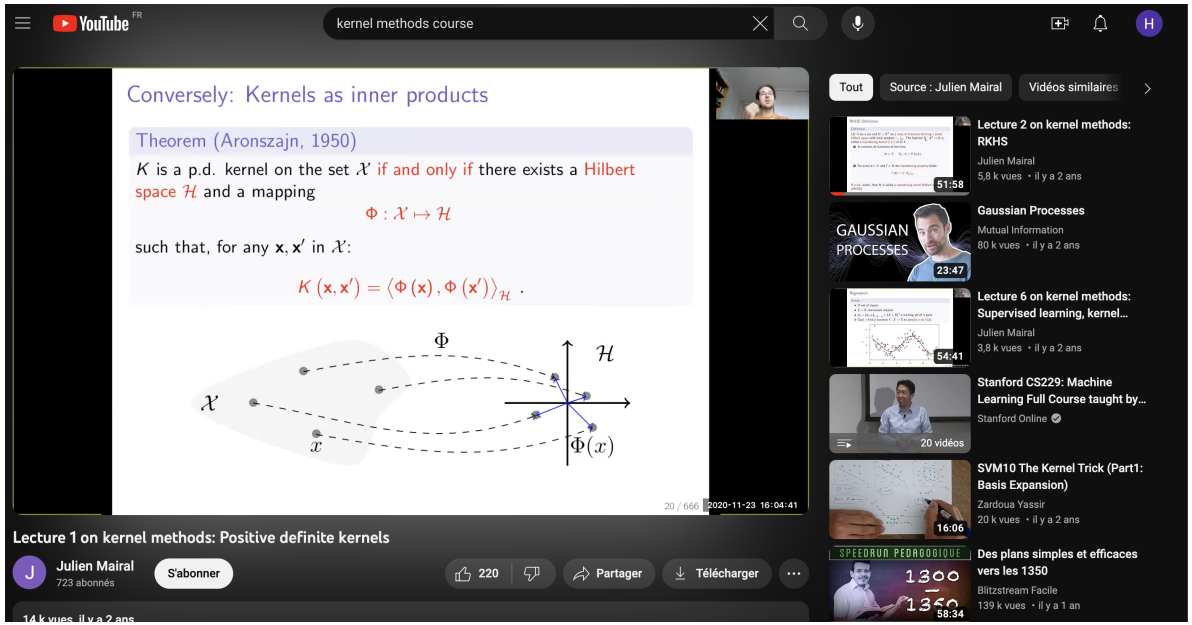
- Need for **continuous**-action methods

10 to 100 million auctions per second in all Criteo servers!

- Need for **efficient** and **scalable** learning methods

# Example 2: YouTube Recommendation

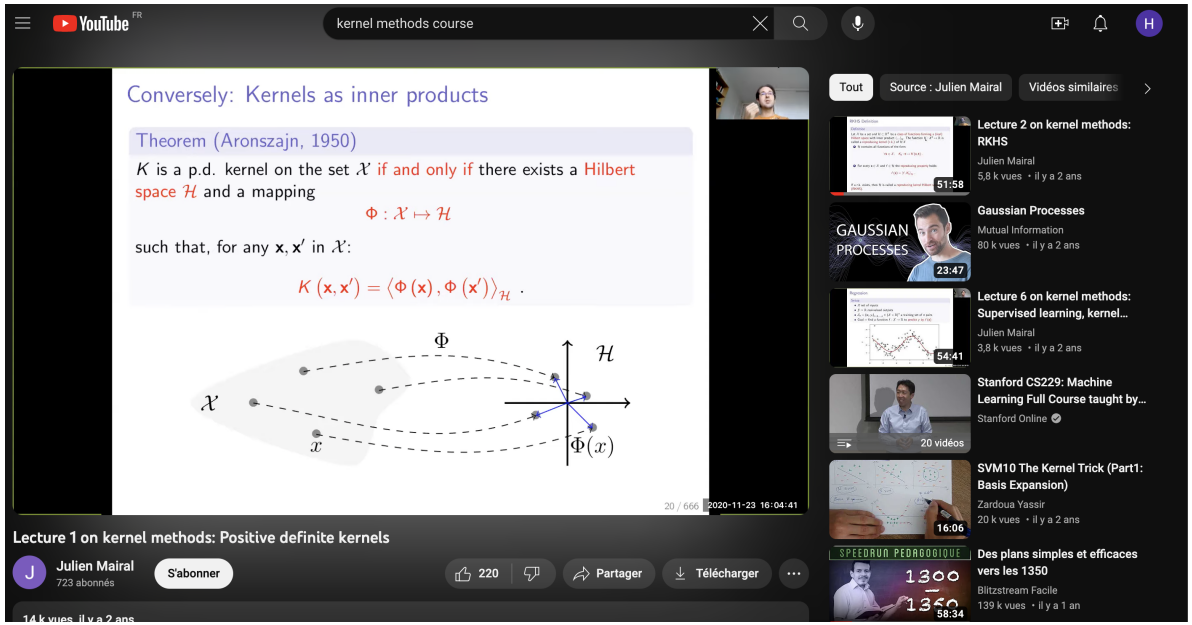
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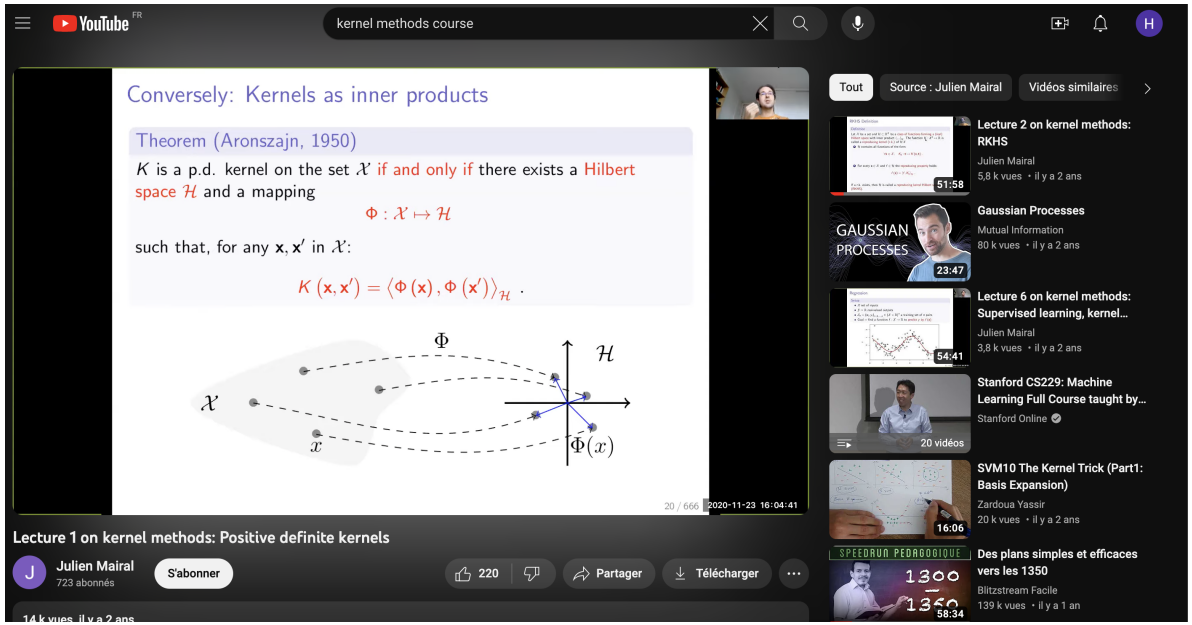



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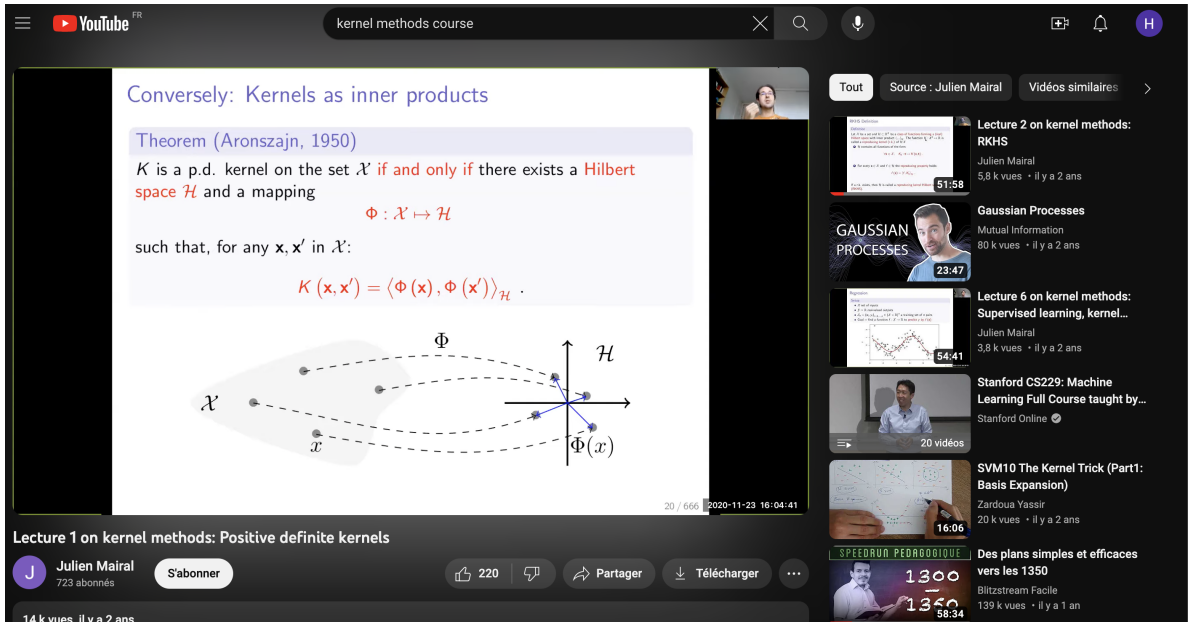



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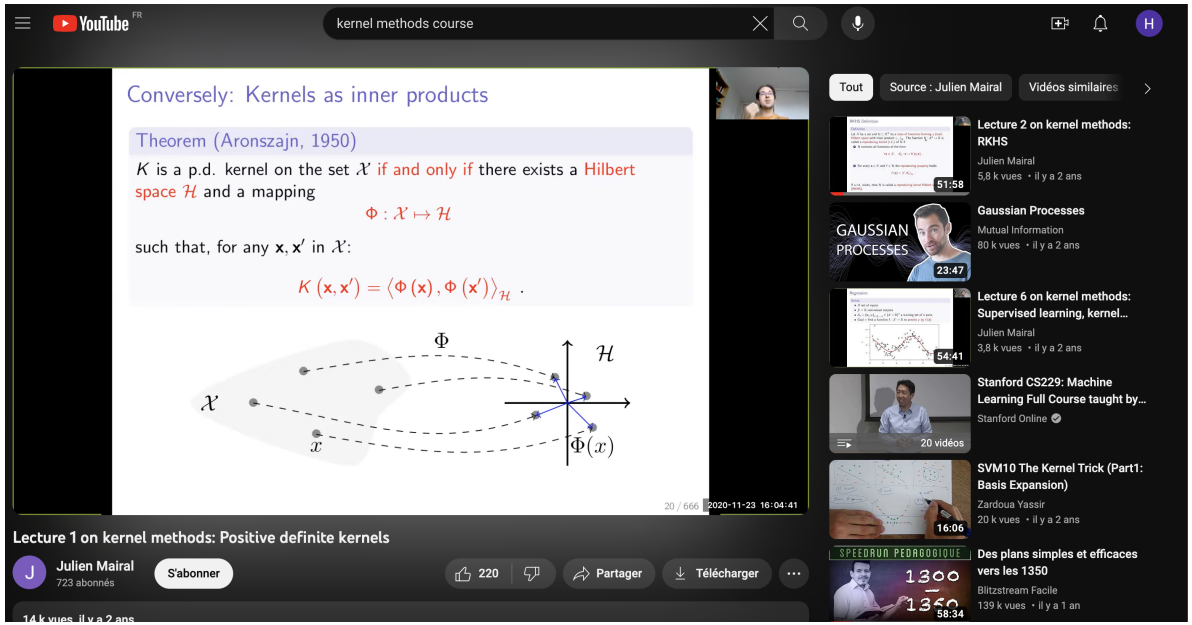



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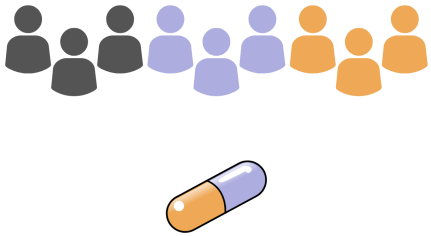
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800 millions videos on YouTube, estimated 1 billion hours watched per day.

- **Large amounts** of log data

# Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



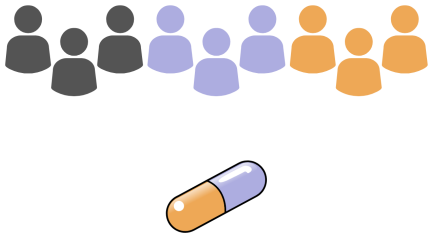
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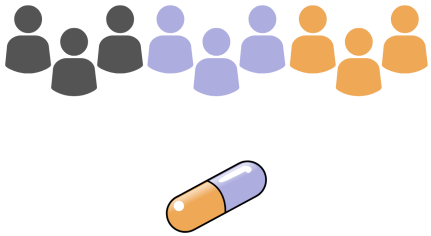


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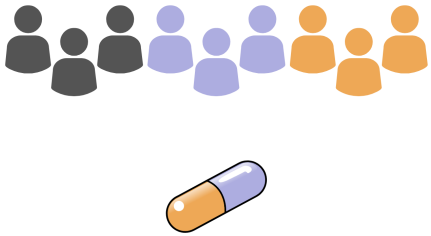


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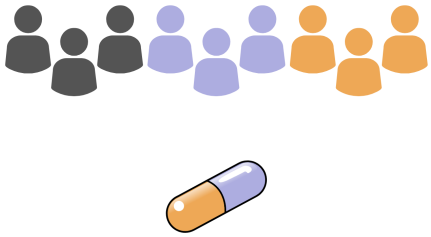


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Hard to hire patients for clinical trials, exploration can be unethical.

- Need for **adaptive** experiments and **safe** methods.

*Disclaimer: personalized medicine is often not used in real-world medical applications because of legal issues related to exploration and safety.*

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Focus on **effectiveness** and **efficiency** of the methods.

# Part I - Efficient methods in sequential learning

Contributions

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# Contextual Bandits

At each round $t$, a bandit **agent** interacts with the environment:

- observes a **context** $x_t \in \mathcal{X}$,
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The goal is to find an **agent** (policy) that **minimizes** the **regret**, over any horizon $T$:

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Well known methods include $\epsilon$-greedy, RegCB, SquareCB (Dylan J. Foster, 2020; Naoki Abe, 1999), and more simply Thompson sampling and OFUL-learning methods (Abbasi-yadkori et al., 2011).

# Kernel UCB

In the Kernel UCB setting, we assume

$$r_t = \langle \theta^*, \phi(x_t, a_t) \rangle + \epsilon_t \,,$$

where $\theta^* \in \mathcal{H}$ the RKHS, $\phi$ is a feature map associated to $\mathcal{H}$ and the kernel $k$.

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The algorithm estimates $\theta^*$ with a regularized least squares:

$$\hat{\theta}_t \in \underset{\theta \in \mathcal{H}}{\arg\min} \left\{ \sum_{s=1}^{t} \left( \langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

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The algorithm takes the upper bound on $\mathcal{C}_t$ and then selects the **best action**:

$$\text{K-UCB}_t(a) = \max_{\theta \in \mathcal{C}_t} \langle \theta, \phi(x_t, a) \rangle, \quad a_t \in \arg\max_{a \in \mathcal{A}} \text{K-UCB}_t(a)$$

**Proposition (K-UCB regret (Zenati et al., 2022))**

*The algorithm enjoys the following pseudo-regret bound:*

$$R_T \lesssim \sqrt{T}\left(\|\theta^*\|\sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}}\right),$$

$d_{\text{eff}}$ is the effective dimension associated to $\lambda$ and the kernel matrix $K_t = [\langle \phi(x_i, a_i), \phi(x_j, a_j)\rangle]_{1\leq i,j\leq t}$.

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$\hat{\mu}_{t,a} \leftarrow K_{\mathcal{S}_{t-1}}(s_{t,a})^\top (K_t + \lambda I_t)^{-1} Y_{t-1}$

$\hat{\sigma}^2_{t,a} \leftarrow \frac{1}{\lambda} k(s_{t,a}, s_{t,a}) - \frac{1}{\lambda} K_{\mathcal{S}_{t-1}}(s_{t,a})^\top (K_t + \lambda I_t)^{-1} K_{\mathcal{S}_{t-1}}(s_{t,a})$

where $s_{t,a} = (x_t, a)$, and let $K_t(s) = [k((x_1, a_1), s), \dots k((x_t, a_t), s)]^\top$

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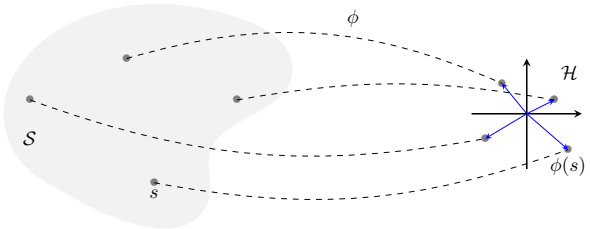
$\hat{\sigma}^2_{t,a} \leftarrow \frac{1}{\lambda} k(s_{t,a}, s_{t,a}) - \frac{1}{\lambda} K_{\mathcal{S}_{t-1}}(s_{t,a})^\top \ (K_t + \lambda I_t)^{-1} \ K_{\mathcal{S}_{t-1}}(s_{t,a})$

where $s_{t,a} = (x_t, a)$, and let $K_t(s) = [k((x_1, a_1), s), \dots k((x_t, a_t), s)]^\top$

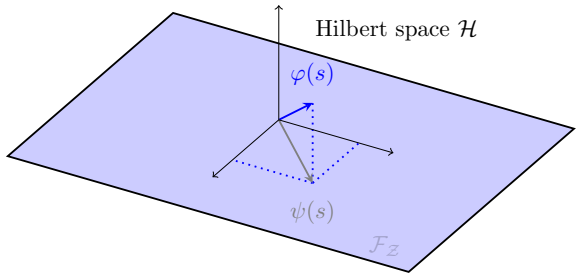
**Computational complexity**

The algorithm runs in $\mathcal{O}(T^3)$ **time complexity**.

# Incremental approximations of the RKHS (Calandriello et al., 2017)




Embedding in the RKHS $\mathcal{H}$

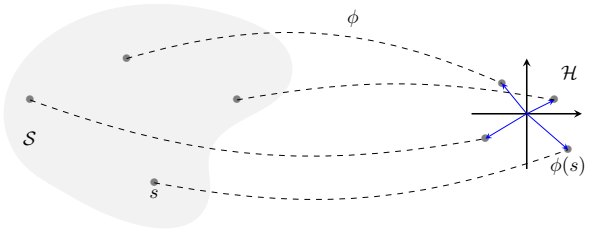



Nyström projection of a feature map $\varphi$

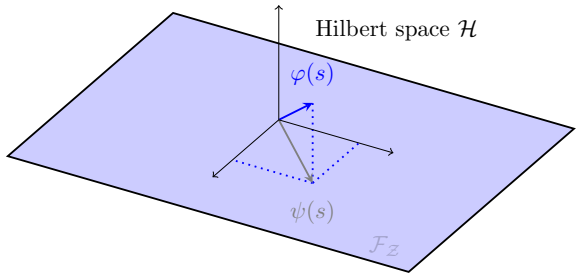
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---

**Algorithm** Incremental KORS subroutine

---

**Input:** Past dictionary $\mathcal{Z}$, context-action $s_t$, regularization $\mu$, budget $\gamma$
Compute the leverage score $\tilde{\tau}_t$ from $\mathcal{Z}, s_t, \mu, \epsilon$
  Compute $\tilde{p}_t = \min\{\gamma \tilde{\tau}_t, 1\}$
  Draw $z_t \sim \mathcal{B}(\tilde{p}_t)$ and if $z_t = 1$, add $s_t$ to $\mathcal{Z}$

---

**Proposition**

*The sequence of dictionaries $\mathcal{Z}_1 \subset \mathcal{Z}_2 \subset \ldots \mathcal{Z}_T$ learned by KORS with parameters $\mu > 0$ satisfies with high probability that $m = |\mathcal{Z}_t| \approx d_{\mathrm{eff}}(\mu, T)$.*

# Efficient Kernel UCB regret bound (Zenati et al., 2022)

EK-UCB estimates the parameter $\theta^*$ with a **projected** parameter $\tilde{\theta}_t$ to build a confidence set $\tilde{\mathcal{C}}_t$:

$$\tilde{\theta}_t \in \underset{\theta \in \mathcal{F}_{\mathcal{Z}}}{\arg\min} \left\{ \sum_{s=1}^{t} \left( \langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

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**Theorem (EK-UCB regret)**

*Writing $m := |\mathcal{Z}_T|$, the regret of the EK-UCB algorithm matches*

$$R_T \lesssim \sqrt{T} \left( \sqrt{\frac{\mu m}{\lambda}} + \sqrt{d_{\text{eff}}} \right) \left( \|\theta^*\| (\sqrt{\lambda} + \sqrt{\mu}) + \sqrt{d_{\text{eff}}} \right).$$

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In the update rule, we now only invert a matrix of size $m \times m$, with $m \approx d_{\text{eff}}$.

**Computational complexity**

The algorithm runs in $\mathcal{O}(T d_{\text{eff}}^2)$ **time complexity**.

# Related Work

Previous works BKB (Calandriello et al., 2019) and BBKB (Calandriello et al., 2020) considered a **non contextual** setting.

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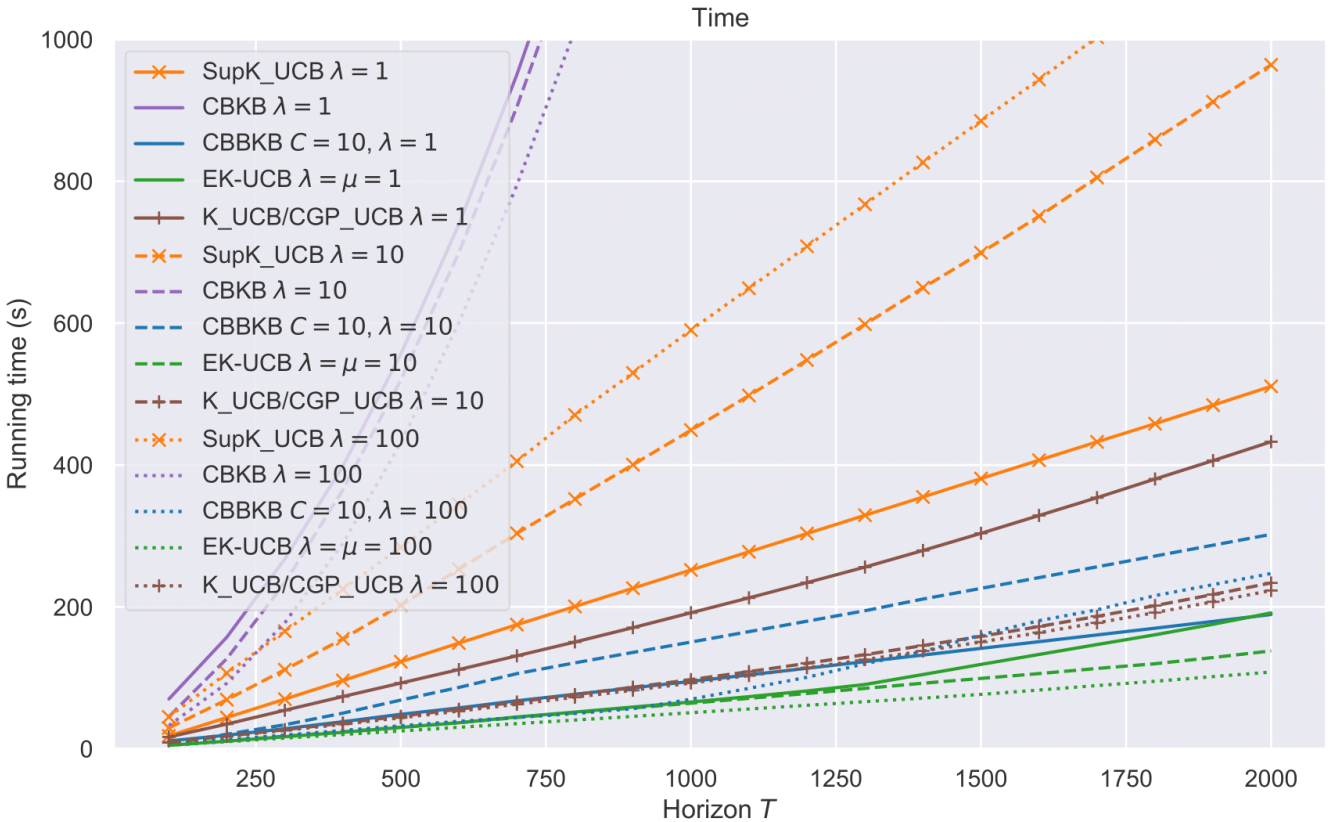
| Algorithm | Time Complexity |
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| CGP-UCB (Krause & Ong, 2011) | $\mathcal{O}(T^3)$ |
| SupKernelUCB (Valko et al., 2013) | $\mathcal{O}(T^3)$ |
| K-UCB (ours) | $\mathcal{O}(T^3)$ |
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Comparison of regrets, space and time complexities

$\implies$ The incremental projection updates allow to perform **more efficient** approximations in the joint context-action space.

# Numerical Experiments (Zenati et al., 2022)

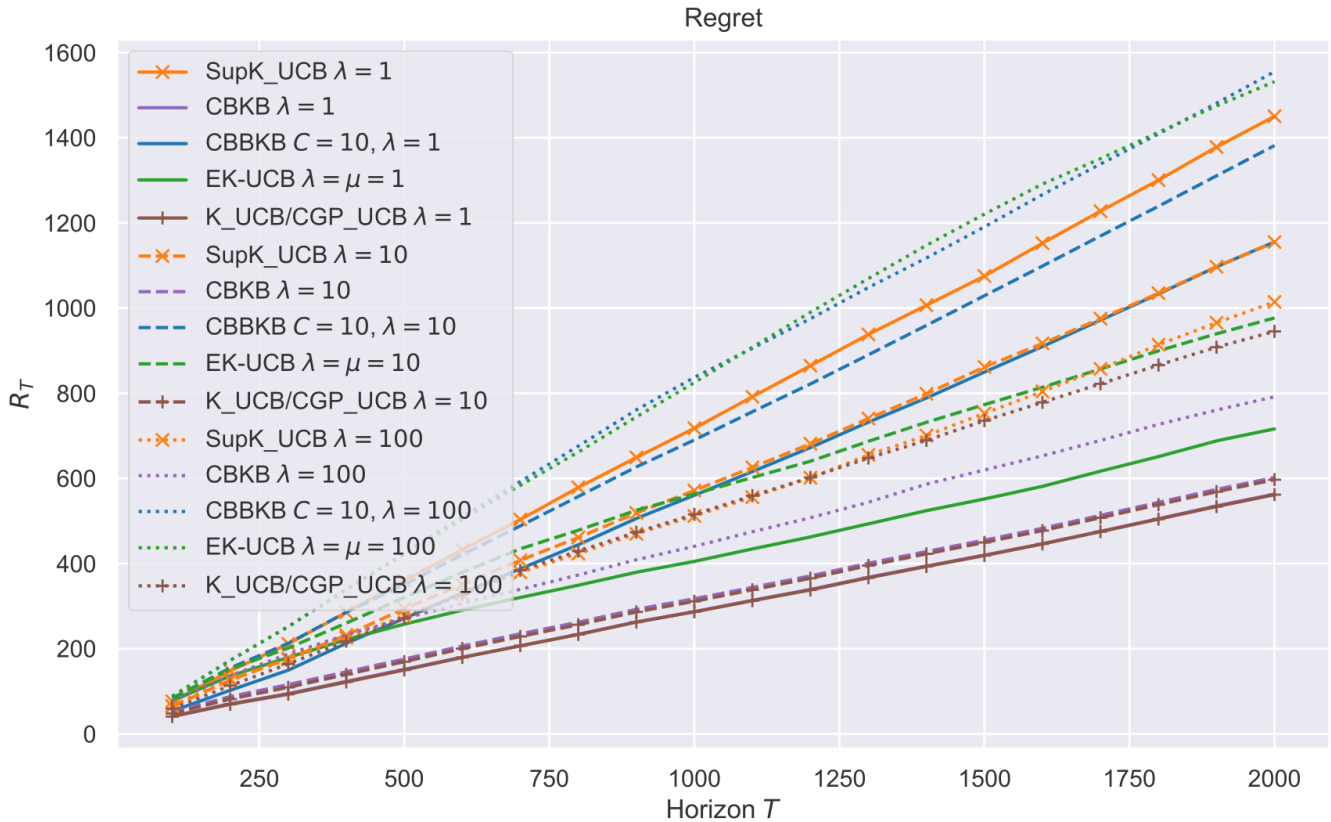
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- The practical computational complexity of "efficient" BKB can be worse than the original algorithm!
- EK-UCB with parameters $\mu = 1, 10$ have low regret.

# Discussions

- The EK-UCB algorithm runs in $\mathcal{O}(Td_{\text{eff}}^2)$ time complexity, which significantly improves over the standard contextual kernel UCB method.
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In some applications (at Criteo especially), an **adversary** can learn the behavior of the agent and adapt their strategy.

What if we consider that setting?

# Another work: nested bandits

In sequential decision making, consider the adversarial setting with $k$ arms and a horizon $T$: Exponential Weights algorithms (Littlestone & Warmuth, 1994) suffer:

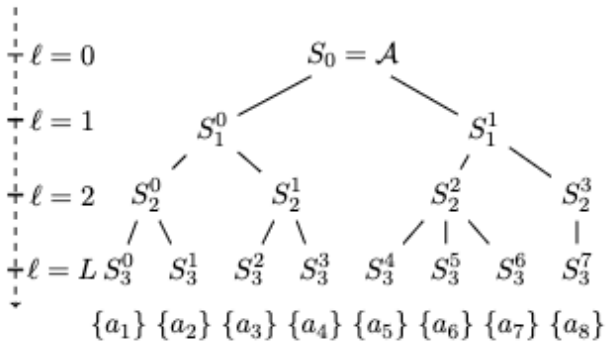
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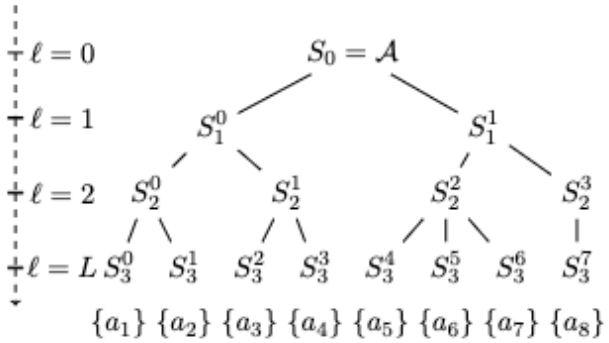

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"Nested" similarity structure

We propose **Nested Exponential Weights**:

$$R_T \leq \sqrt{2k_{\text{eff}} \log(k) T} \quad \text{and} \quad k_{\text{eff}} \ll k.$$

Martin, M., Mertikopoulos, P., Rahier, T., & Zenati, H. (2022). Nested bandits. *International Conference on Machine Learning (ICML)*

# Part II - Efficient Counterfactual Learning Methods

## Contributions

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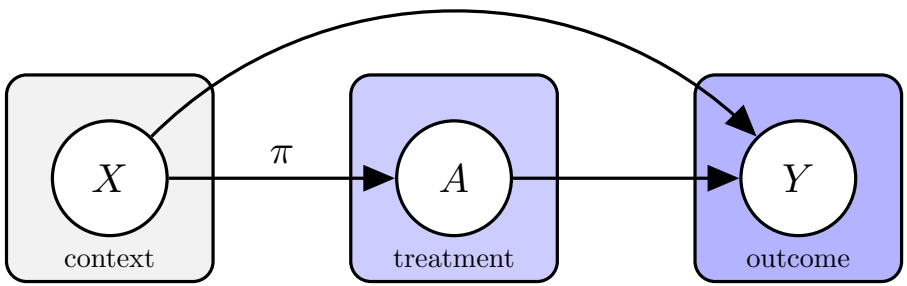
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The goal is to find a policy that **minimizes** the **risk**, the expected loss of running $\pi$:

$$L(\pi) = \mathbb{E}_{x\sim\mathcal{P}_{\mathcal{X}}}\, \mathbb{E}_{a\sim\pi(\cdot|x)}\, \mathbb{E}_{y\sim\mathcal{P}_{\mathcal{Y}}(\cdot|x,a)}\, [y]$$

This requires off-policy evaluation (modelling + estimation) and optimization.

# Counterfactual reasoning

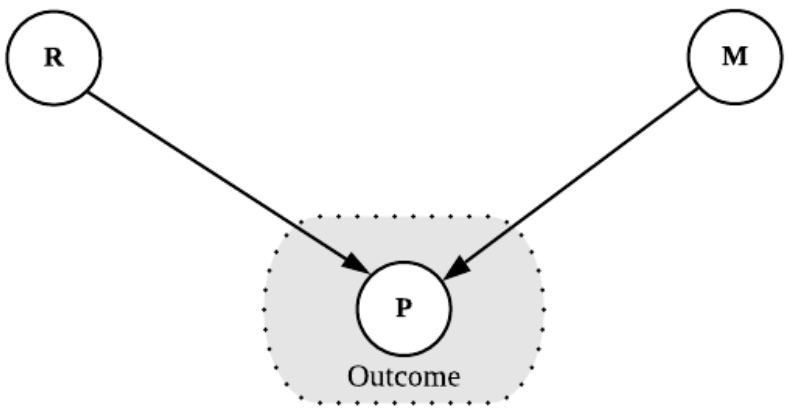
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Example problem: *"At INRIA Grenoble the majority of PhD students often hikes in the mountains. They end up graduating a PhD. What if they did not hike? Would they still graduate?"*.

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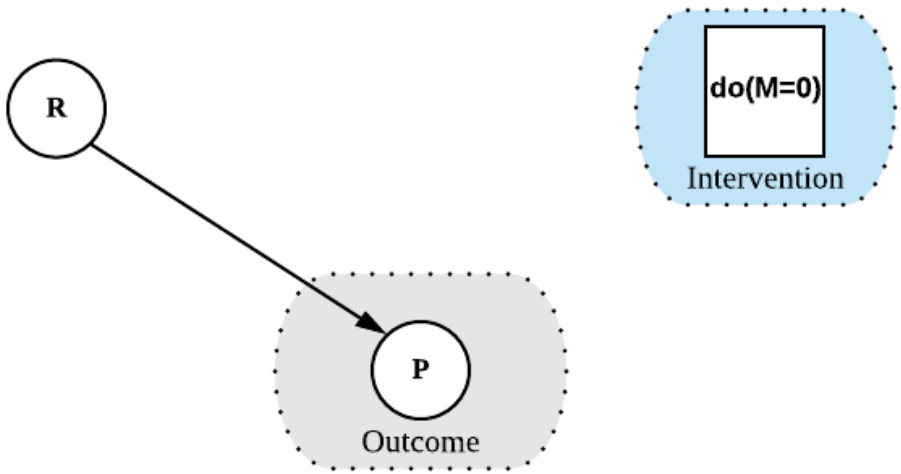


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The difficulty lies in making a prediction about **unobserved cases** based only on **observed information**.

We consider **parametric** policies $\pi_\theta$ associated to a **model** $\theta \in \Theta$. The risk $L$ of a **model** $\theta \in \Theta$ is given by:

$$L(\theta) = \mathbb{E}_{x \sim \mathcal{P}_{\mathcal{X}}} \mathbb{E}_{a \sim \pi_\theta(\cdot|x)} \mathbb{E}_{y \sim \mathcal{P}_{\mathcal{Y}}(\cdot|x,a)} [y]$$

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By using the IPS method (Horvitz & Thompson, 1952):

$$L(\theta) = \mathbb{E}_{a\sim\pi_\theta(\cdot|x),x,y}\, [y] = \mathbb{E}_{a\sim\pi_{\theta_0}(\cdot|x),x,y}\, \left[ y \frac{\pi_\theta(a|x)}{\pi_{\theta_0}(a|x)} \right]$$

Then, a natural estimator is

$$\hat{L}^{\text{IPS}}(\theta) = \frac{1}{n} \sum_{i=1}^{n} y_i \frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)}. \qquad \text{(IPS)}$$

Importance weighting has large variance, a clipped estimator has been proposed:

$$\hat{L}^{\text{cIPS}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{y_i \min\left\{\frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)}, \alpha\right\}}_{\nu_i(\theta)}, \qquad \text{(cIPS)}$$

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**Counterfactual Risk Minimization** (CRM) Swaminathan and Joachims (2015) find a model $\theta$ that minimizes a regularized estimate of $L(\theta)$.

$$\hat{\theta}^{\text{CRM}} \in \arg\min_{\theta\in\Theta} \hat{L}^{\text{cIPS}}(\theta) + \lambda\sqrt{\frac{\widehat{\text{Var}}(\nu_i(\theta))}{n}} \quad \text{with} \quad \lambda > 0. \qquad \text{(CRM)}$$

The learning objective is conservative.

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- Self-normalized estimator (SNIPS) (Owen, 2013):

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- Doubly robust estimator (DR) (Dudik et al., 2014):

$$\hat{L}^{\text{DR}}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\eta}(x_i, a_i))\,\frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)} + \frac{1}{n}\sum_{i=1}^{n}\sum_{a\in\mathcal{A}}\hat{\eta}(x_i, a)\pi_\theta(a|x_i), \tag{DR}$$

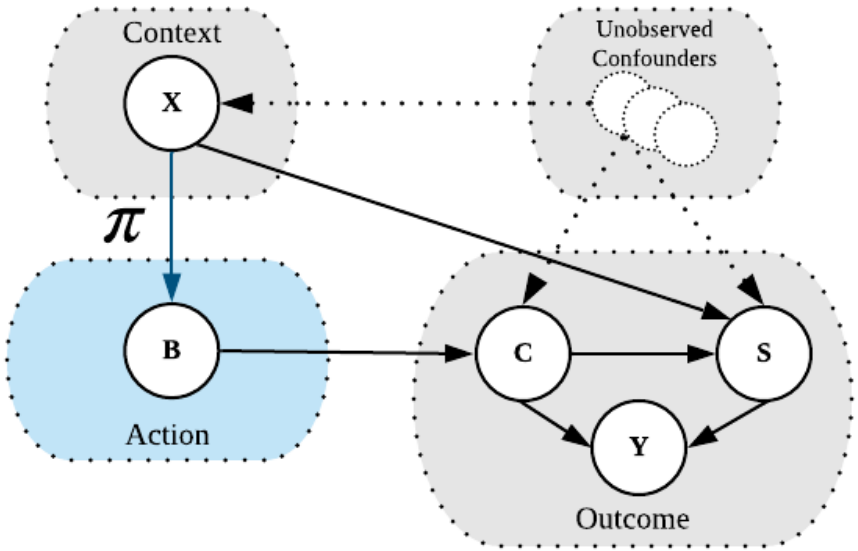
with a learned estimator $\hat{\eta}(x, a)$ of expected costs, for instance by using ridge regression to fit $y_i \approx \hat{\eta}(x_i, a_i)$ on the training data.

# Continuous action problems, a new dataset: CoCoA (Zenati et al., 2020)

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CoCoA causal graph

$B$ is **bid**. The **cost** $c \sim C$ and a number of **sales** $S$ defines a **loss** $Y$.

| $n$ | $d$ | $\mathbb{E}[-Y]$ | $\mathbb{V}[Y]$ | $\mathbb{V}[A]$ | $\mathbb{P}(Y \neq 0)$ |
|---|---|---|---|---|---|
| $120.10^6$ | 3 | 11.34 | 9455 | .01 | .07 |

CoCoA dataset summary statistics.

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- **Diagnose** when the offline estimation is too uncertain: use the effective sample size (Owen, 2013).
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Still, it remains **difficult** to obtain results in real-world problems because the methods often fail to pass the offline evaluation protocol:

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Test rewards of continuous action methods: the higher the better

If the logging policy did not explore well the action space, offline learning becomes very difficult!

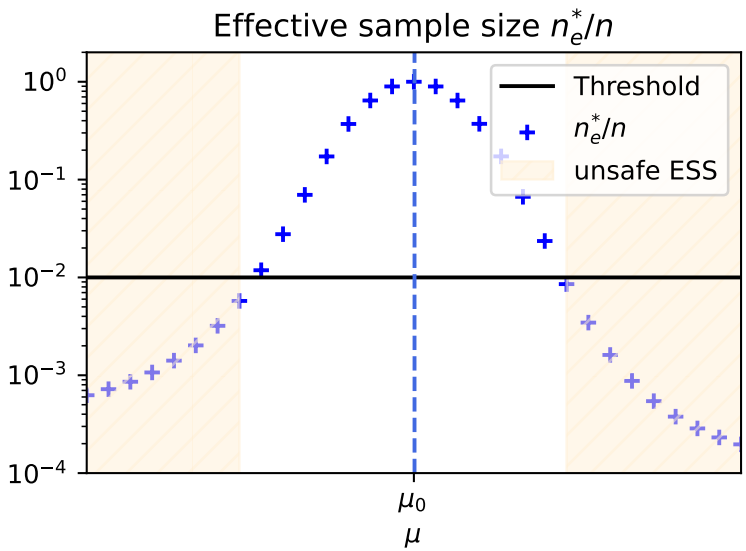
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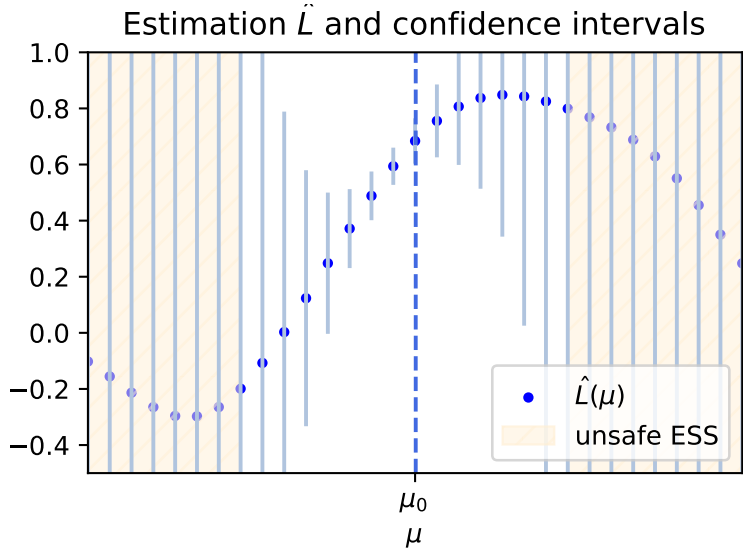
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Consider Gaussian parametrized policies $\pi_\mu = \mathcal{N}(\mu, \sigma^2)$. We evaluate $L(\mu)$ based on samples drawn from a logging policy associated to $\mu_0$.

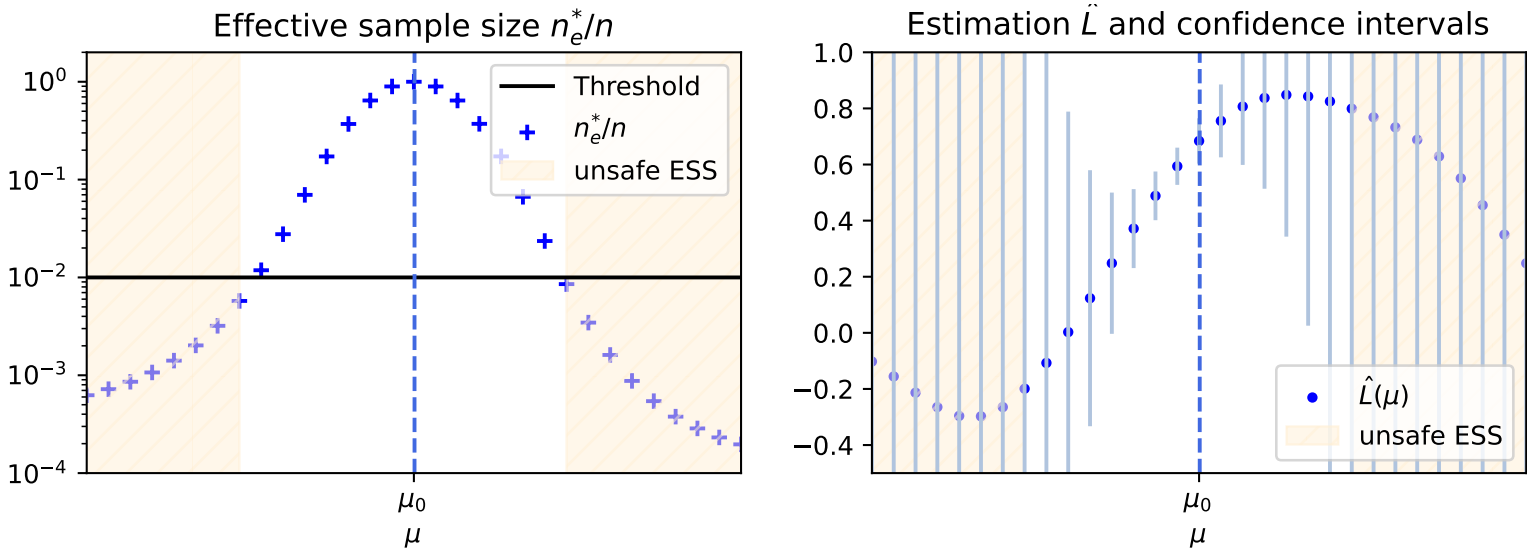


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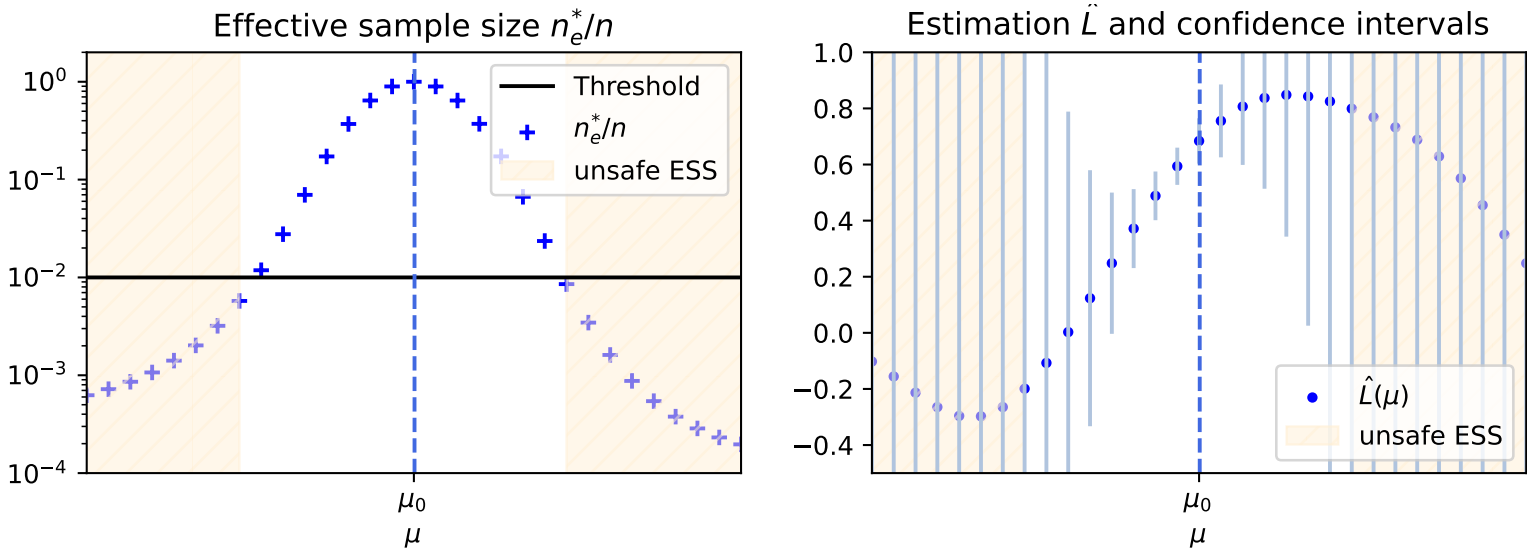



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$\Longrightarrow$ effective sample sizes can be too low and confidence intervals too large.

Therefore it is sometimes necessary to increase the sample size by collecting additional samples with **sequential** deployments.

$\Longrightarrow$ **Adaptively** collect data instead of **redeploying** the same logging model $\theta_0$.

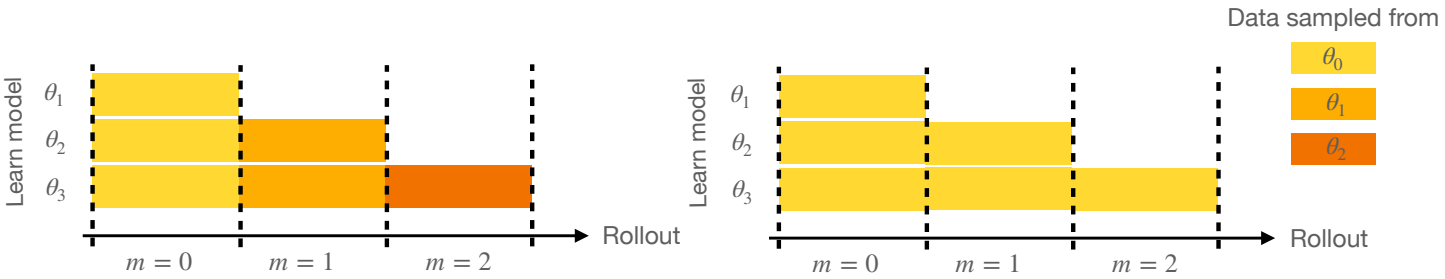



**Figure:** Sequential designs (left) and non-adaptive redeployment (right).

At round $m$, a model $\theta_m \in \Theta$ is deployed and observations $s_m = (x_{m,i}, a_{m,i}, y_{m,i})_{i=1,\dots,n_m}$ are collected to learn a new model $\theta_{m+1}$.

---

**Algorithm** Sequential Counterfactual Risk Minimization

---

**Input:** Logged observations $(x_{0,i}, a_{0,i}, y_{0,i}, \pi_{0,i})_{i=1,\dots,n_0}$, parameter $\lambda > 0$

**for** $m = 0$ *to* $M - 1$ **do**

  Build $\mathcal{L}_m$ from observations $s_m$

  Learn $\theta_{m+1}$

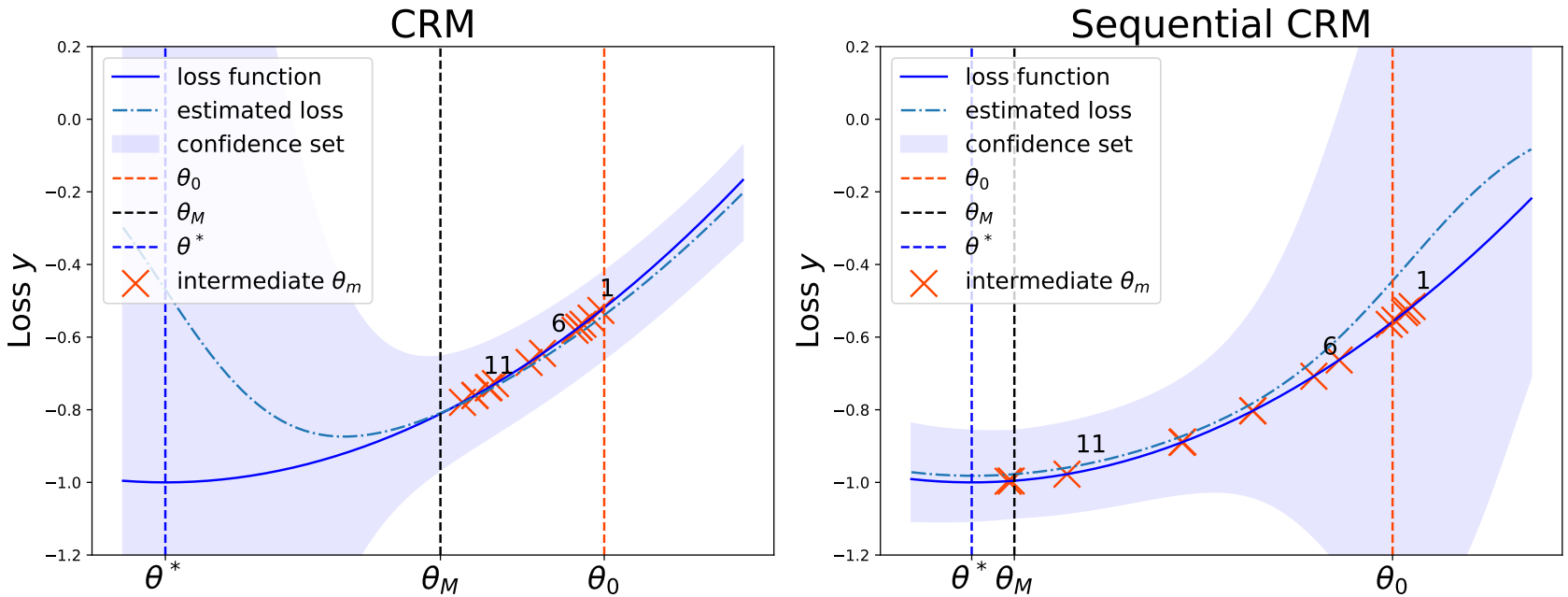
  Deploy the model $\theta_{m+1}$ and collect observations $s_{m+1}$

**end**

---

# Illustrating the benefits of sequential designs (Zenati et al., 2023)

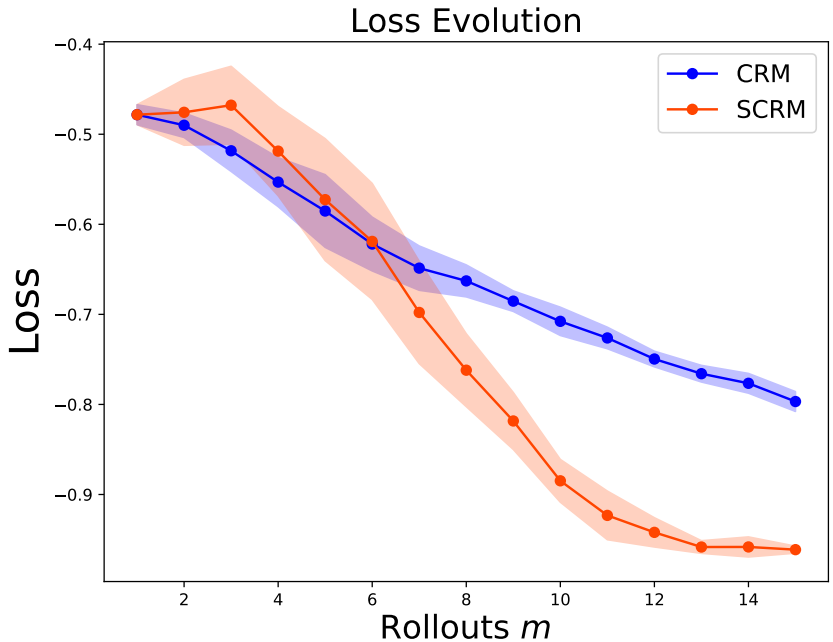
Consider Gaussian parametrized policies $\pi_\theta = \mathcal{N}(\theta, \sigma^2)$ with quadratic losses $l_t(a) = (a - y_t)^2 - 1$ where $y_t \sim \mathcal{N}(\theta^*, \sigma^2)$.




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# IPS-IX: Implicit exploration and controlled variance (Zenati et al., 2023)

We propose a new counterfactual estimator. Denoting $\pi_{m,i} = \pi_{\theta_m}(a_{m,i}|x_{m,i})$ and $\alpha > 0$, we change the importance weighted estimator:

$$\begin{cases} \nu_{i,m}(\theta) = y_{m,i} \min\left\{ \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i}}, \alpha \right\} & \hat{L}^{\text{cIPS}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \nu_{i,m}(\theta) \qquad \text{(cIPS)} \\ \zeta_{i,m}(\theta) = y_{m,i} \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i} + \alpha \pi_{\theta,i}} & \hat{L}^{\text{IPS-IX}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \zeta_{i,m}(\theta) \qquad \text{(IPS-IX)} \end{cases}$$

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**Proposition (Generalization Error Bound, informal)**

*Let $\theta \in \Theta$, then with high propability:*

$$L(\theta) \lesssim \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{Var}(\zeta_{i,m}(\theta) - y_{m,i})}{n_m}} + \frac{\lambda_m^2}{n_m},$$

*for a $\lambda_m$ depending on a complexity measure of $\Theta$.*

IPS-IX resembles the implicit exploration estimator in the EXP3-IX algorithm (Lattimore & Szepesvári, 2020).

The latter motivates to solve for $m \in \{1, \dots M\}$:

$$\hat{\theta}_{m+1} \in \arg\min_{\theta \in \Theta} \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{\text{Var}}(\zeta_{i,m}(\theta) - y_{m,i})}{n_m}}. \qquad \text{(SCRM)}$$

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**Proposition (Excess Risk Bound, informal)**

*With high probability:*

$$L(\theta_{m+1}) - L(\theta^*) \lesssim \sqrt{\nu_m^2 \frac{\log n_m}{n_m}},$$

*where* $\nu_m^2 = \text{Var}_{x,\theta_m}\left(\frac{\pi_{\theta^*}(a|x)}{\pi_{\theta_m}(a|x)}\right)$.

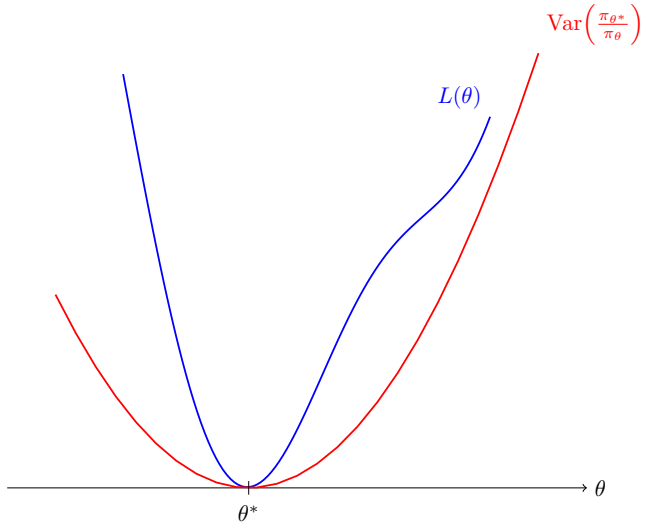
$\implies$ IPS-IX improves the dependency in $\nu_m^2$, compared to $\nu_m^2 + 1$ in previous work (Zenati et al., 2020).

$\implies$ crucial for using these error bounds **sequentially** since $\nu_m \to 0$ if $\theta_m \to \theta^*$.

**Assumption (Hölderian Error Bound)**

*There exist* $\gamma > 0$ *and* $\beta > 0$ *such that for any* $\theta \in \Theta$,

$$\gamma \mathrm{Var}_{x,\theta}\left(\frac{\pi_{\theta^*}(x|a)}{\pi_\theta(x|a)}\right) \leq (L(\theta) - L(\theta^*))^\beta .$$




Classical bounds are of the form $\gamma d(\theta, \theta^*) \leq (L(\theta) - L(\theta^*))^\beta$ with $d$ a distance.

**Proposition (SCRM Excess Risk, informal)**

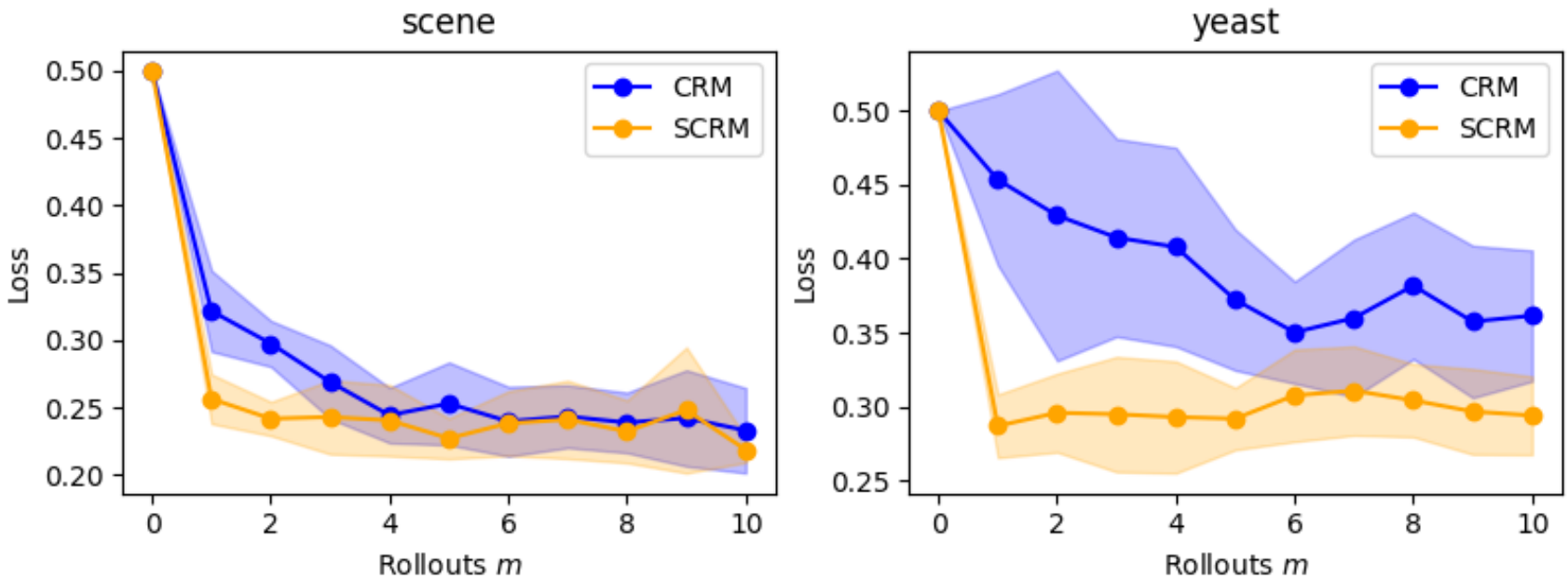
*Let* $n_0, n \geq 2$, **geometric** *sample sizes* $n_m = n_0 2^m$ *for* $m = 0, \ldots, M \approx \left\lfloor \log \frac{n}{n_0} \right\rfloor$.

$$L(\theta_M) - L(\theta^*) \leq O\left(n^{-\frac{1}{2-\beta}} \log n\right).$$

Fast rates on par with ERM (Bartlett & Mendelson, 2006; Van Erven et al., 2015) and when $\beta = 0$, we recover the CRM rates (Zenati et al., 2020).

# SCRM in practice (Zenati et al., 2023)

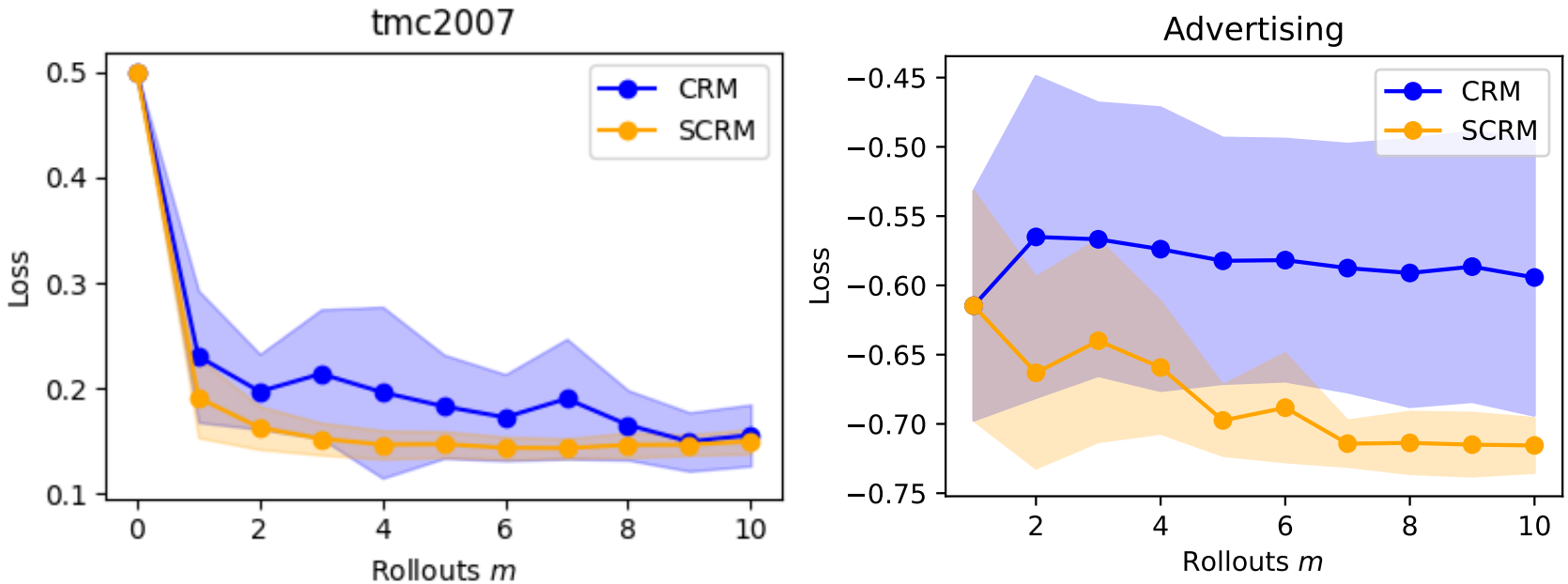
We consider discrete action real-world datasets (*Scene*, *Yeast*, *TMC2007*) and a continuous action setting (*Advertising*).




SCRM converges very fast, often within the first rollouts. Conversely, CRM needs more samples and the variance is higher.

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# Perspectives

- Counterfactual policy learning in real-world applications requires an offline evaluation protocol.
- When the logging policy underexplores the action space, counterfactual learning is hard, instead **sequential** deployments (when possible) can drastically improve the learning.
- We note that the learning objective was **conservative** and it may not encourage **exploration**.

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Examples at Criteo: recommendation, ressource allocation, hyperparameter search.

We list the following future research directions:

**Counterfactual learning methods**

- Investigate the use of doubly robust estimators and other control variate estimators for continuous action methods.
- In SCRM, develop guarantees to use all past samples (particularly effective in practice). To avoid introducing dependencies in the excess risk bounds, we used geometric sample sizes to discard previous samples.

# Perspectives and research directions

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**Contextual bandits**

- In the stochastic setting, make a large benchmark of elimination algorithms to see their practical performance, then investigate the use of efficient methods.
- In the adversarial setting with nesting, extend to contextual information with less assumptions on feedbacks.

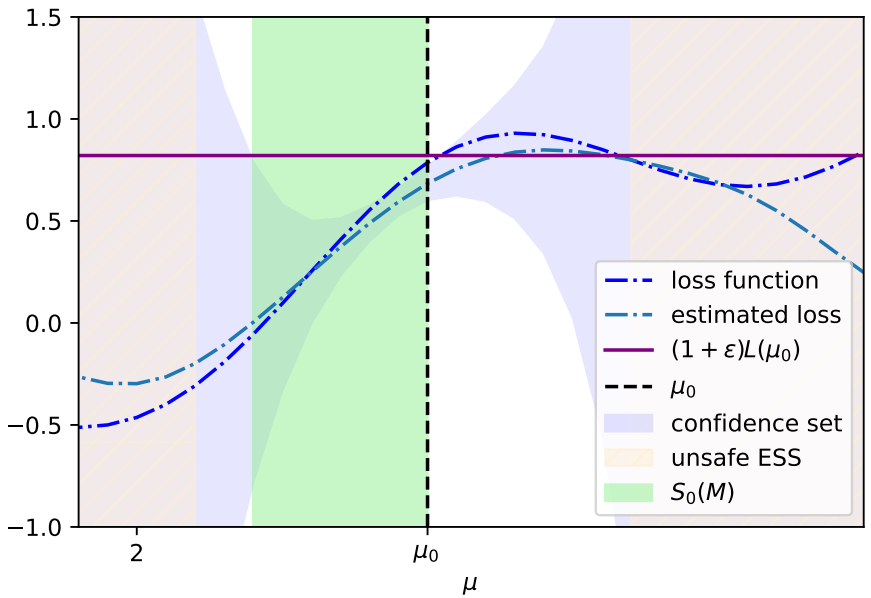
# A promising research direction: safe SCRM

Regarding the exploration-exploitation tradeoff, investigate the optimism in the face of uncertainty (OFUL) principle in SCRM along with safety constraints:

$$\text{LCB}(\theta) = \hat{L}(\theta) + \lambda\sqrt{\frac{\hat{V}(\theta)}{n}} + \frac{\lambda^2}{(n-1)}.$$

We could define a safety set of $\theta \in \Theta$ where the CRM solution lies:

$$S_0(\Theta) = \left\{\theta \in M, \text{ s.t } LCB(\theta) \leq (1+\epsilon)\hat{L}(\theta_0)\right\}.$$



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- Offline counterfactual learning strategies can be preferred in that case but it is **hard** to do especially when the logging system **underexplored** the action space.
- Sequential counterfactual learning can arguably bridge the gap and take the **best of both worlds**.

**Acknowledgment**

- Supervisors: Eustache Diemert, Pierre Gaillard, Julien Mairal
- Co-authors: Alberto Bietti, Matthieu Martin, Panagiotis Mertikopoulos Thibaud Rahier
- My colleagues at Criteo and INRIA
- PhD jury members: Massih-Reza Amini, Olivier Cappé, Nicolò Cesa-Bianchi, Claire Vernade

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